From tests to models

[...] much of the open science movement has been focused on misconduct and questionable-research-practices (like p-hacking), instead of research procedures like ***data management and justification of modeling choices****.*

We need to define the phenomenon, the alternative explanations, and what estimates would help us ***distinguish among them or refine them****.*

However, the biggest problem with Bayes factors is none of these. Rather it is that people want to use them to do null hypothesis testing, and testing null hypotheses is the fuel of the garbage fire in the first place. ***Cumulative science needs non-null scientific models and ways to confront them with data****.*

Richard McElreath

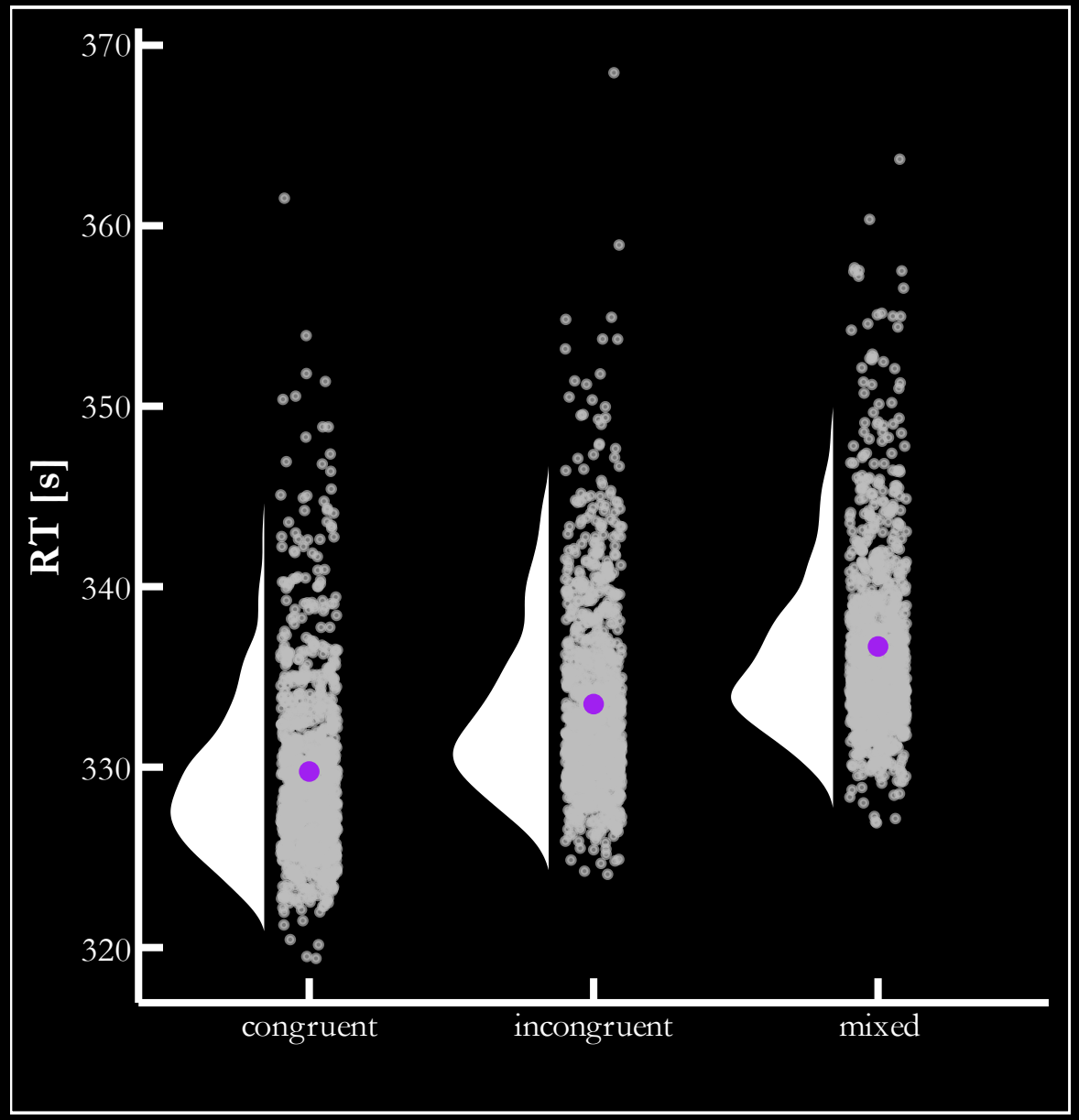
Objectives

- Demystifying data modelling
- Showing that modelling is more understandable than tests
- Showing that we often model without knowing it
- Moving to Bayesian models

Approach

- Unpacking a simple ANOVA
- Showing that ANOVA is a linear model* in disguise
- Running a first Bayesian model

* Technically is a method applied to a linear model



Are the means different?

How can we answer this question?

Frequentist	ANOVA Linear Moel
Bayesian	General Linear Model

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Argument:

a) ANOVA and Linear Model are equivalent
b) Linear models are more flexible than standard tests
c) Bayesian models offer more information than frequentists linear models
d) Then: we should opt for Bayesian models

Steps

1. Load data
2. Compute participants' and conditions' averages
3. Compute Variance Metrics
4. Get F-statistics
5. Working on p-value

NOTE: The code needs to be adjusted (do not rush through)

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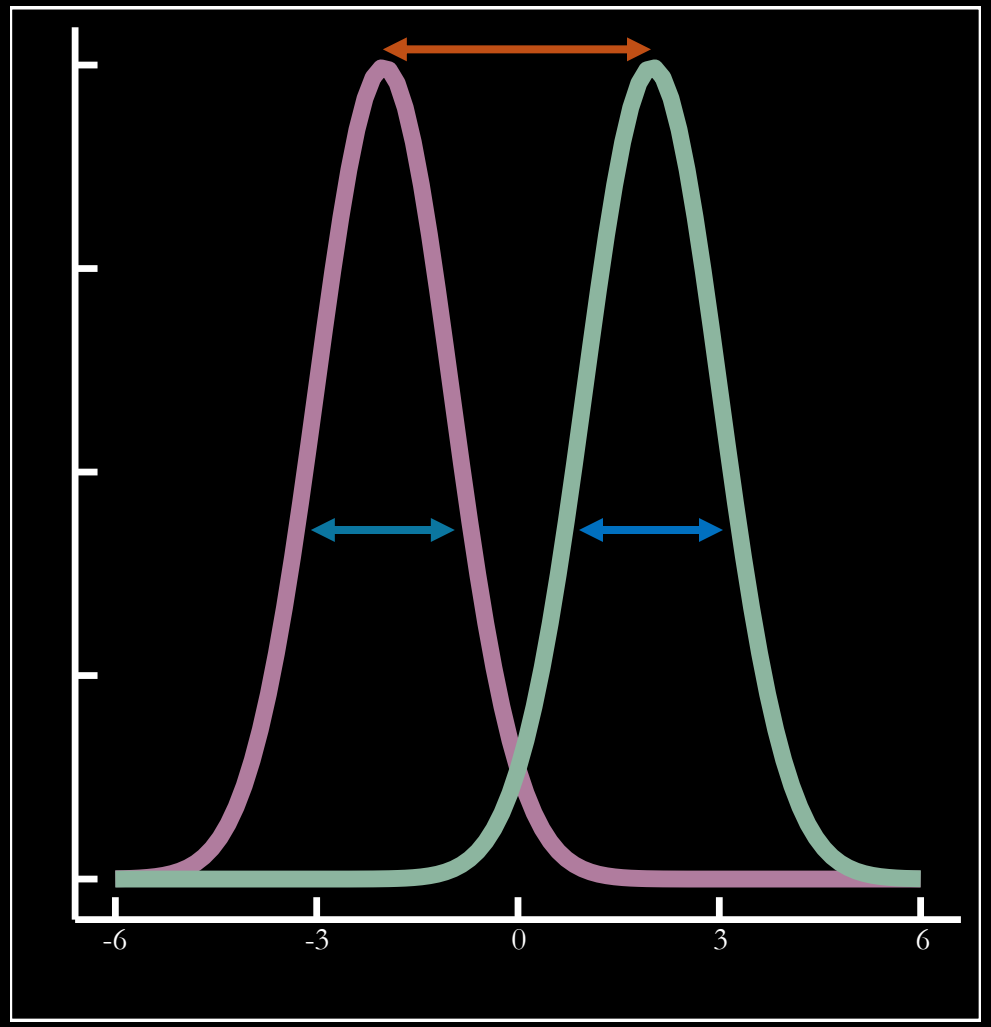

If variance between groups is higher than variance within groups, then the observations are likely drawn from two different distributions

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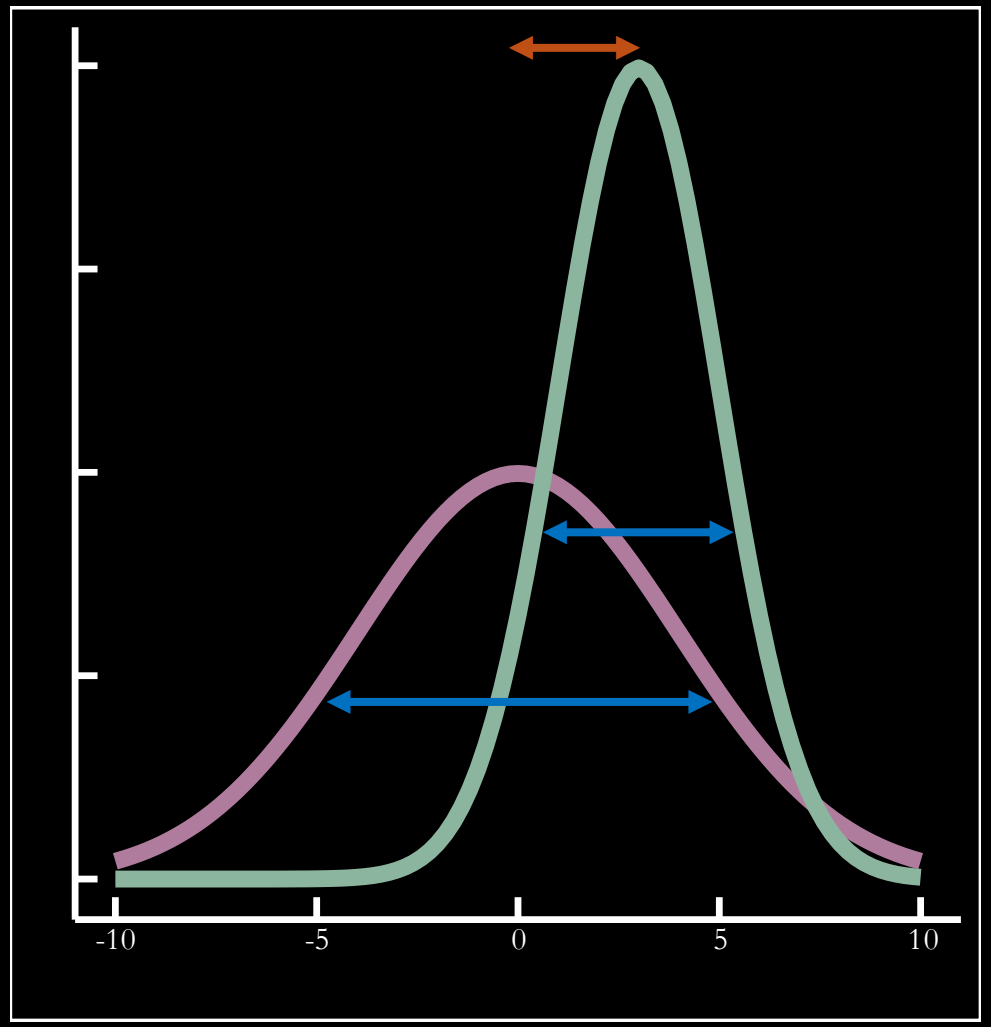

Otherwise, it's **hard to tell**

NOTE: This has implications for interpreting a null result!

F-value

$$F = \frac{Between\ Group\ Variability}{Within\ Group\ Variability} = \frac{\frac{SS_{between}}{Df_{between}}}{\frac{SS_{within}}{Df_{within}}} = \frac{SS_{between}}{Df_{between}} \times \frac{Df_{within}}{SS_{within}} = \frac{SS_{between}}{SS_{within}} \times \frac{Df_{within}}{Df_{between}}$$

SS = Sum of Squares

In case

$$SS_{within} = \sum (\overline{RT}_p - \overline{RT}_{g_p})^2$$

$$SS_{between} = \sum N_g \times (\overline{RT}_g - \overline{RT})^2$$

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ANOVA assumption!

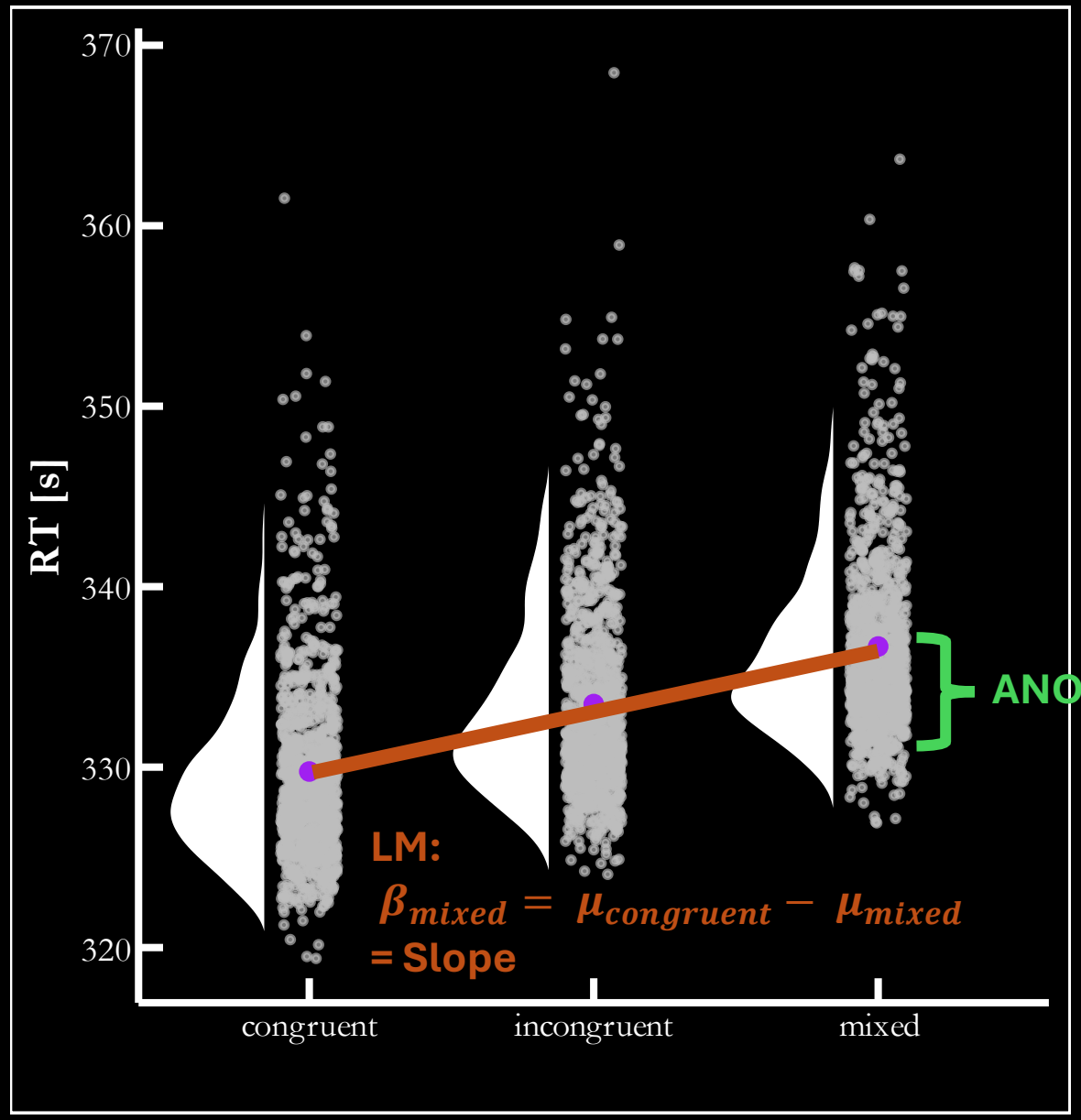
Linear Model

- H0 looks like a linear model, right?
  Indeed, ANOVA is a method **applied to** a linear model
- *F* has some similarities with R^2

$$F = \frac{SS_{between}}{SS_{within}} \times \frac{Df_{within}}{Df_{between}}$$

$$R^2 = 1 - \frac{SS_{residuals}}{SS_{total}}$$

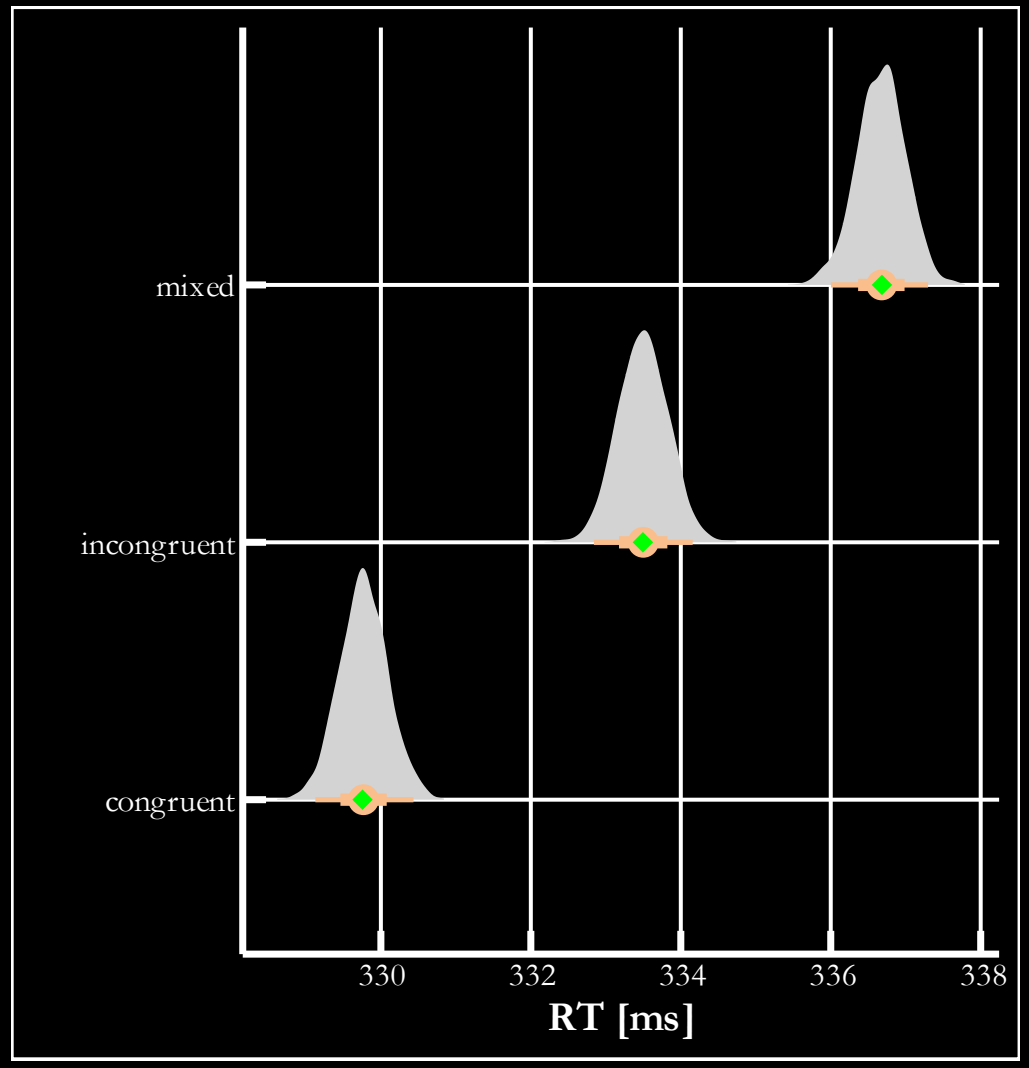
Proportion of variance in the outcome captured by the predictor


370
360
350
340
330
320
RT [s]
congruent
incongruent
mixed
LM:
$\beta_{mixed} = \mu_{congruent} - \mu_{mixed}$
= Slope
ANOVA: $\mu_{congruent} - \mu_{mixed}$

Going Bayesian

- A linear model is a more flexible tool than and ANOVA (or any other tests)
- The switch between the two it’s quite easy
- The switch to a Bayesian approach is quite easy too
  - In terms of code
  - As for anything, some aspects require learning and adjusting, BUT…
  - The benefits in terms of interpretability and general practice outweigh the challenges

Going Bayesian

- Both Frequentist and Bayesian approaches use likelihoods
  - We'll see the similarity through code

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